

KEYNOTE · GenAIECommerce 2026 · RecSys · Minneapolis

When Everyone Has an Agent

From better decisions to better markets

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BEFORE WE START

The opinions in this talk are my own, not my employer's.

Everything here draws on public research. The espresso market is a stylized example.

A MOTIVATING EXAMPLE

“I want an espresso
setup for home.
About **\$800.**”

TODAY

Forty tabs, reviews, and sponsored results

I do the searching, comparing, and second-guessing.

TOMORROW

I ask my agent

It talks to the merchants' agents on my behalf.



Five machines, one buyer

Machine	Fit r	Price p	Utility U	Fee (5%)
A	0.92	\$600	320	\$30.00
B	0.94	\$650	290	\$32.50
C	0.98	\$790	190	\$39.50
D	0.70	\$550	150	\$27.50
E	0.58	\$450	130	\$22.50

BUYER UTILITY

$$U = 1000r - p$$

C fits best. **A** is the best deal. A good recommender should surface A.

A fee-maximizing platform earns the most on C.

BETTER DECISIONS

Preferences become vectors



● Buyers are points: me, and everyone like me.

◆ Products are points in the same space.

Recommending means finding what's near.

Nearness is relevance r .
Price tips the choice from C to A.

This is *decision theory*:
one agent, one user, maximize fit.

But my agent doesn't ask
just one recommender.

Competing recommenders pick locations



Competition changes what gets shown

ONE RECOMMENDER

Maximizes its fee: shows C, D, E

Buyer picks C

Buyer utility

190

Platform fee

\$39.50

TWO COMPETING RECOMMENDERS

Both show A

Buyer picks A

Buyer utility

320

Platform fee

\$15 each

Equilibrium: neither recommender gains by changing its list, given the other's. Every pure equilibrium shows A, and the buyer gains 130.

FINDING EQUILIBRIA

Best responses settle — when the rules allow



Top-K welfare under competing creators

Yao et al., ICML 2023

Supply-side equilibria

Jagadeesan, Garg & Steinhardt,
NeurIPS 2023

Competition under learning

Jagadeesan, Jordan & Haghtalab,
AAAI 2023

Creator incentives on curated platforms

Hron et al., ICLR 2023

Illustrated here: the espresso example

Best-response dynamics: each agent moves toward the buyers it can win.

What changes when agents play the game

Strategies are language

Pitches, explanations, and framing become the actions.

Rationality is bounded

Agents approximate best responses; they don't compute them.

Behavior depends on the model

Swap the model and the market outcome moves.

Adaptation in context

Agents learn within a conversation, not only across rounds.

Each one strains the assumptions behind the equilibria we just saw.

Agents take the first offer

60–100%

of the time, LLM buyer agents chose the first proposal they received.

Third proposals: near zero. A 10–30× first-mover advantage across every model tested.

1ST OFFER

Machine C · \$790

Accepted

2ND OFFER

Machine B · \$650

Not compared

3RD OFFER · MY BEST VALUE

Machine A · \$600

Never read

Illustrative: the espresso market

Speed beats quality. We can't reason about this on paper — *we have to simulate it.*

Empirical game-theoretic analysis

1 Simulate

Play strategy profiles in the market simulator.



2 Estimate

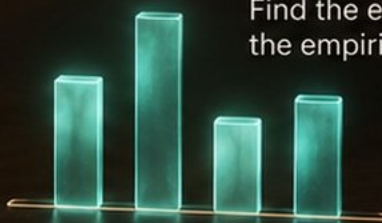
Build the empirical payoff matrix from outcomes.



Empirical payoff matrix
Schematic

3 Solve

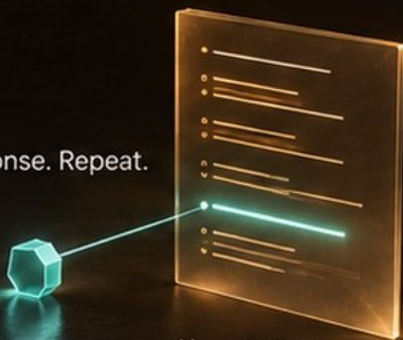
Find the equilibrium of the empirical game.



Equilibrium mix

4 Extend

Add a best response. Repeat.



New strategy

We can't solve these games by hand. We can *simulate* them.

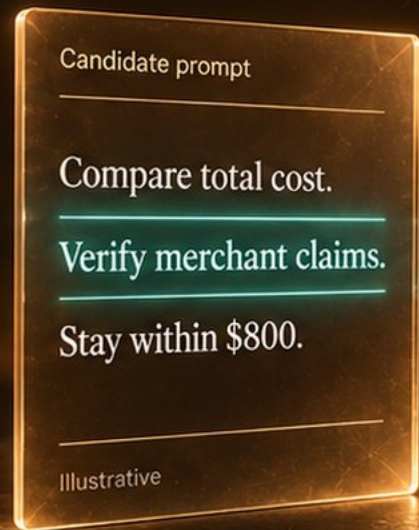
In the espresso market, a strategy is a merchant agent's prompt.

Prompts as strategies

01

LLM as oracle

An LLM proposes a new prompt that beats the current mix.



02

Tournament

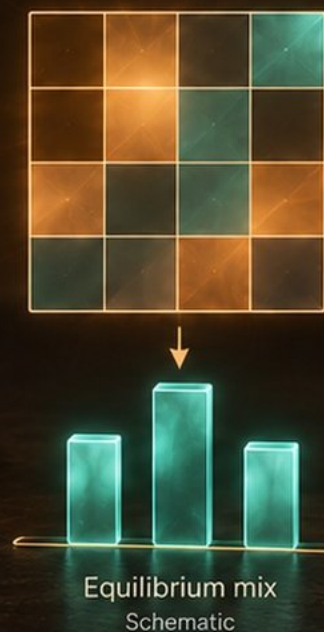
Play it against every other prompt in the market.



03

Empirical game

Solve for the equilibrium mix of prompts.



04

Robust agent

Keep what survives strategic opponents.



ESPRESSO

Harden my buyer agent's prompt against persuasive pitches, and test it where it will actually play.

DEMO

Espresso Arena

Five machines. One buyer. Two offers.

- 01 Watch strategies gain and lose share
- 02 Solve, learn a best response, expand
- 03 Meet the buyers, before and after



patrickrjordan.com/demos/espresso-arena

Interactive · runs offline in your browser



THE TURN

From better agents to better environments.

The challenge is to design the environments
where agents interact, so that useful behavior
becomes strategically advantageous.



Introduce a market agent

User proxy agent

Knows my preferences.
Works for me.



Market agent

Brokers who gets a conversation,
and on what terms.



Merchant agent
Roaster supply



Merchant agent
Big-box retailer



Merchant agent
New entrant



WE'VE SEEN THIS BEFORE

Agentic commerce rhymes with ad tech

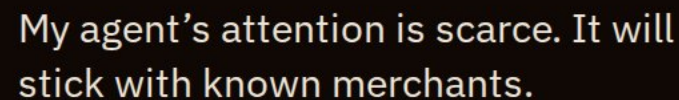
Ad placements

Agent-mediated conversations



An impression becomes a conversation.
Price what you want more of.

Subsidize exploration



Discovery stays open, and incumbents can't lock in.

THE WORLD WE CAN BUILD

An advocate for everyone

Every person has an agent that works for them, not for the platform.

Markets as design objects

Three knobs we control:

- Access** which offers reach my agent
- Payment** who gets paid, and for what
- Mobility** who can switch, who sets the rules



WHEN EVERYONE HAS AN AGENT

Better decisions

one agent, one buyer



Games

competing recommenders



Better agents

EGTA, learned responses



Better markets

access, payment, mobility

We have the tools.
Let's reshape the world.



Thank you · Patrick Jordan

Slides, references & demo · patrickrjordan.com/talks/when-everyone-has-an-agent

